

Wen-Chi (Vicky) Tsai

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WORK EXPERIENCE

Research Assistant, Agent Harness and Model Co-Optimization Sep. 2026 – Present
Language Technologies Institute, CMU, advised by Prof. Graham Neubig Pittsburgh, PA

- Built a **failure-analysis pipeline over 720 agent trials** of raw JSON telemetry, classifying each trial by exit state and exposing a **15.4% silent failure rate** the reported mean absorbed, a 10-point AIME swing
- Selected evaluation subsets with **nearest-neighbour search over a 1.79M-row dataset**, re-embedding all **17,398 DAPO-Math problems** with sentence-transformers to rule out a text artifact, 91 of 100 picks surviving
- Closed a **silent cross-job failure on a shared multi-tenant GPU cluster**, where a dead vLLM server sent every trial to another job's endpoint, with dynamic port allocation and served-model assertions **shipped as a pull request**

Software Engineering Intern Jun. 2025 – Mar. 2026
ASUS, Open Cloud Infrastructure Software R&D Team Taipei, Taiwan

- Cut a full regression pass from **~16 hours to ~1–2 hours** with a **TypeScript end-to-end (E2E) test suite** covering ~95% of previously manual cases across Linux, Windows, and macOS
- Automated issue triage with a **GitLab Model Context Protocol (MCP) agent**, pairing an LLM classification prompt with MCP tool calls to auto-classify, label, and route issues
- Shipped four **context providers into a VS Code AI coding assistant**, adding commit-history, OS-metadata, current-file, and repository-map context selected from an evaluation of candidates

Startup Engineer Jun. 2026 – Aug. 2026
RoomMuse Pittsburgh, PA

- Caught unbuildable layouts at design time rather than on install day with a **six-rule geometric conflict engine** that blocks any unit sealing a socket, pipe, or doorway while passing intended connections (**45 tests**)

Research Assistant, Edge LLM Compression Sep. 2025 – Feb. 2026
Wireless Ad-Hoc and Sensor Networks Laboratory, NCU Taoyuan, Taiwan

- Scaled teacher-student distillation to **data-parallel training on 3 GPUs with DeepSpeed ZeRO-2**, sharding optimizer state and gradients, since **ZeRO-3 parameter sharding corrupts GPTQ-packed weights**
- Recovered **4-bit accuracy at 2-bit model size** (IoU 0.9998, 1.18 GB) by fine-tuning a Llama-3.2 student with **GPTQ quantization and LLM-Neo knowledge distillation** in PyTorch
- Surveyed **llama.cpp, vLLM, and SGLang** for serving on a Jetson AGX Orin edge target, mapping **latency versus throughput** across PagedAttention, continuous batching, and prefix-cache reuse

SELECTED PROJECTS

Dual-Strategy UAV Path Planning Sep. 2024 – Dec. 2025

- Designed a dual-strategy planner trading **layered MST-DFS** for structured-space traversal against **collision-aware A* replanning**, cutting path-generation time over $5\times$ and path length up to 14%
- Built the **3D point-cloud pipeline in C++ and Python**, slicing raw .pcd scans into layers and chaining **per-layer KNN and MST** into a DFS traversal, with normal-vector correction for full coverage

Museum Artifact Recognition App May 2024 – Nov. 2024

- Shipped **on-device model inference in an iOS app**, running the team's YOLOv8n and Pix2Pix reflection-removal models in **Swift with TensorFlow Lite** at 94.5% accuracy across 7 classes

PUBLICATIONS

Wen-Chi Tsai, Yi-Peng Liu, Yu Wu, Yun-Sheng Chen, Chung-Lun Lai, Yu-Yu Chang, Min-Te Sun. "UAV Path Planning for Building Structure Inspection." In *14th International Conference on Ubi-Media Computing (Ubi-Media 2026)*, Penang, Malaysia.

EDUCATION

Carnegie Mellon University (CMU), School of Computer Science Aug. 2026 – Dec. 2027 (Expected)
Master of Science in Intelligent Information Systems, Language Technologies Institute Pittsburgh, PA

- Relevant coursework:** AI Agents, Multimodal Machine Learning, Intro to Question Answering with LLMs

National Central University (NCU) Sep. 2022 – Jun. 2026
Bachelor of Science in Computer Science and Information Engineering | GPA: 3.99/4.30 Taoyuan, Taiwan

- Relevant coursework:** Data Structures, Algorithmics, Operating Systems, Computer Networks, Software Engineering